Chapter 15, Problem 55P

Problem

Solve for *y*(*t*) in the following differential equation if the initial conditions are zero.

$$\frac{d^3y}{dt^3}+6\frac{d^2y}{dt^2}+8\frac{dy}{dt}=e^{-t}\cos 2t$$

Step-by-step solution

Step 1 of 6

$$\frac{d^3y}{dt^3}+\frac{6d^2y}{dt^2}+8\frac{dy}{dt}=e^{-t}\cos 2t$$

Applying transforms and equating initial conditions to zero.

$$s^3Y(s)-s^2y(0)-sy'(0)-y''(0)+6\left[sY(s)-sy(0)-y'(0)\right]+8\left[sY(s)-y(0)\right]=\frac{s+1}{(s+1)^2+4}$$

$$Y(s)\left[s^3+6s^2+8s\right]=\frac{s+1}{(s+1)^2+4}$$

$$Y(s)=\frac{s+1}{s\left[s^2+6s+8\right]\left[(s+1)^2+4\right]}$$

Step 2 of 6

$$Y(s)=\frac{s+1}{s(s+2)(s+4)(s^2+2s+5)}$$

$$Y(s)=\frac{A}{s}+\frac{B}{s+2}+\frac{C}{s+4}+\frac{Ds+E}{s^2+2s+5}$$

$$A=\left.\frac{s+1}{(s+2)(s+4)(s^2+2s+5)}\right|_{s=0}$$

$$A=\frac{1}{2\times4\times5}$$

$$A=\frac{1}{40}$$

Step 3 of 6

$$B = \left.\frac{s+1}{s(s+4)(s^2+2s+5)}\right|_{s=-2}$$

$$B = \frac{-1}{-2\times 2\times 5}$$

$$B = \frac{1}{20}$$

$$C = \left.\frac{s+1}{s(s+2)(s^2+2s+5)}\right|_{s=-2}$$

$$C = \frac{-3}{-4\times -2\times 13}$$

$$C = \frac{-3}{104}$$

Step 4 of 6

$$\frac{s+1}{s(s+2)(s+4)(s^2+2s+5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+4} + \frac{Ds+E}{s^2+2s+5}$$

$$\begin{aligned} s+1 = {} & A(s+2)(s+4)(s^2+2s+5) + Bs(s+4)(s^2+2s+5) \\ & + Cs(s+2)(s^2+2s+5) + (Ds+E)s(s+2)(s+4) \end{aligned}$$

$$\begin{aligned} s+1 = {} & A(s^4+8s^3+25s^2+46s+40) + B(s^4+6s^3+13s^2+20s) + C(s^4+4s^3+9s^2+10s) \\ & + D(s^4+6s^3+8s^2) + E(s^3+6s^2+8s) \end{aligned}$$

$$\begin{aligned} s+1 = {} & s^4(A+B+C+D) + s^3(8A+6B+4C+6D+E) + s^2(25A+13B+9C+8D+6E) \\ & + s(46A+20B+10C+8E) + 40A \end{aligned}$$

Step 5 of 6

Comparing coefficients of s^4

$$A+B+C+D=0$$

$$D=-A-B-C$$

$$D=-\frac{1}{40}-\frac{1}{20}+\frac{3}{104}$$

$$D=-\frac{3}{65}$$

Step 6 of 6

Comparing coefficients of s

$$46A+20B+10C+8E=1$$

$$E=\frac{1-46A-20B-10C}{8}$$

$$E=\frac{1-46\left(\frac{1}{40}\right)-20\left(\frac{1}{20}\right)-10\left(\frac{-3}{104}\right)}{8}$$

$$E=-\frac{7}{65}$$

$$Y(s)=\frac{\left(\frac{1}{40}\right)}{s}+\frac{\left(\frac{1}{20}\right)}{s+2}+\frac{\left(\frac{-3}{104}\right)}{s+4}+\frac{\left(\frac{-3}{65}\right)s+\left(\frac{-7}{65}\right)}{s^2+2s+5}$$

$$Y(s)=\left(\frac{1}{40}\right)\frac{1}{s}+\left(\frac{1}{20}\right)\frac{1}{(s+2)}+\left(\frac{-3}{104}\right)\frac{1}{(s+4)}+\frac{\left(\frac{-3}{65}\right)s+\left(\frac{-3}{65}\right)+\left(\frac{-4}{65}\right)}{(s+1)^2+4}$$

$$Y(s)=\left(\frac{1}{40}\right)\frac{1}{s}+\left(\frac{1}{20}\right)\frac{1}{(s+2)}+\left(\frac{-3}{104}\right)\frac{1}{(s+4)}+\left(\frac{-3}{65}\right)\frac{(s+1)}{(s+1)^2+2^2}+\left(\frac{-2}{65}\right)\frac{2}{(s+1)^2+2^2}$$

Applying inverse transforms we get

$$\boxed{y(t)=\left(\frac{1}{40}+\frac{1}{20}e^{-2t}-\frac{3}{104}e^{-4t}-\frac{3}{65}e^{-t}\cos 2t-\frac{2}{65}e^{-t}\sin 2t\right)u(t)}$$